

Problem definition

Hate speech detection is useful for online moderation, but models can over-rely on surface lexical cues. Core failure mode: non-hateful comments that mention identity groups may be misclassified as hateful. Examples of difficult cases include:

- ▶ neutral discussion of identity groups;
- ▶ counterspeech;
- ▶ critical but legitimate discourse;
- ▶ satire or quoted hateful language.

Research question: *Can model-error-driven contrastive fine-tuning reduce false positives on identity-related non-hateful text while preserving as much hateful recall as possible?*

Literature Gap

Existing work:

- ▶ **HateXplain** uses human rationales to improve the explainability of hate speech classifiers.
- ▶ Rationale-based training can reduce some identity bias, but it does not directly target identity-related false positives.
- ▶ **Sen et al. (2022)** show that generic counterfactual augmentation can even increase false positives on non-hateful identity-related texts.

Our contribution:

- ▶ We focus directly on identity-related false positives produced by our model.
- ▶ We use attribution scores to identify which identity terms drive these errors.
- ▶ We build targeted contrastive pairs to teach the model hateful intent vs. neutral identity mentions.

Method

1 - Baseline Model

Architecture & Training

- Model: RoBERTa-base (pretrained transformer)
- Task: Binary classification (hateful vs. non-hateful)
- Loss: Cross-entropy | Epochs: 3 | Batch size: 16 | LR: 2×10^{-5}

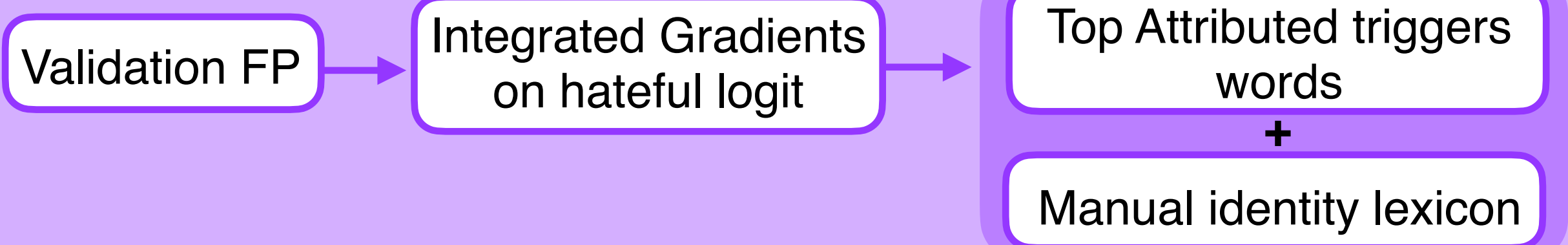
2 - Identify False Positives

Diagnosis Method

- Run inference on validation set
- Collect false positives
- Focus on identity-related false positives as primary issue

FP
gold label = non-hateful
predicted = hateful

3 - Attribution & Candidate Selection



Identity False Positive

We keep validation false positives containing predefined identity terms or highly attributed trigger words.

4 - Manual Pair Construction

Construction Process

- Extract sentence from validation set containing the newly defined identity terms
- Manually write contrastive pairs (non-hateful / hateful examples)
- Both examples preserve identity term, differ in meaning

Quality Guidelines

- Preserve identity term in both versions
- Change semantic meaning, not surface form
- Cross-review and remove unclear examples

5 - Contrastive Fine-Tuning

Mixed-Loss Contrastive Fine-Tuning

L_{CE}^{train} : preserve original task performance

L_{CE}^{pairs} : classify manual paired examples

$L_{contrastive}$: push non-hateful/hateful pair embeddings apart

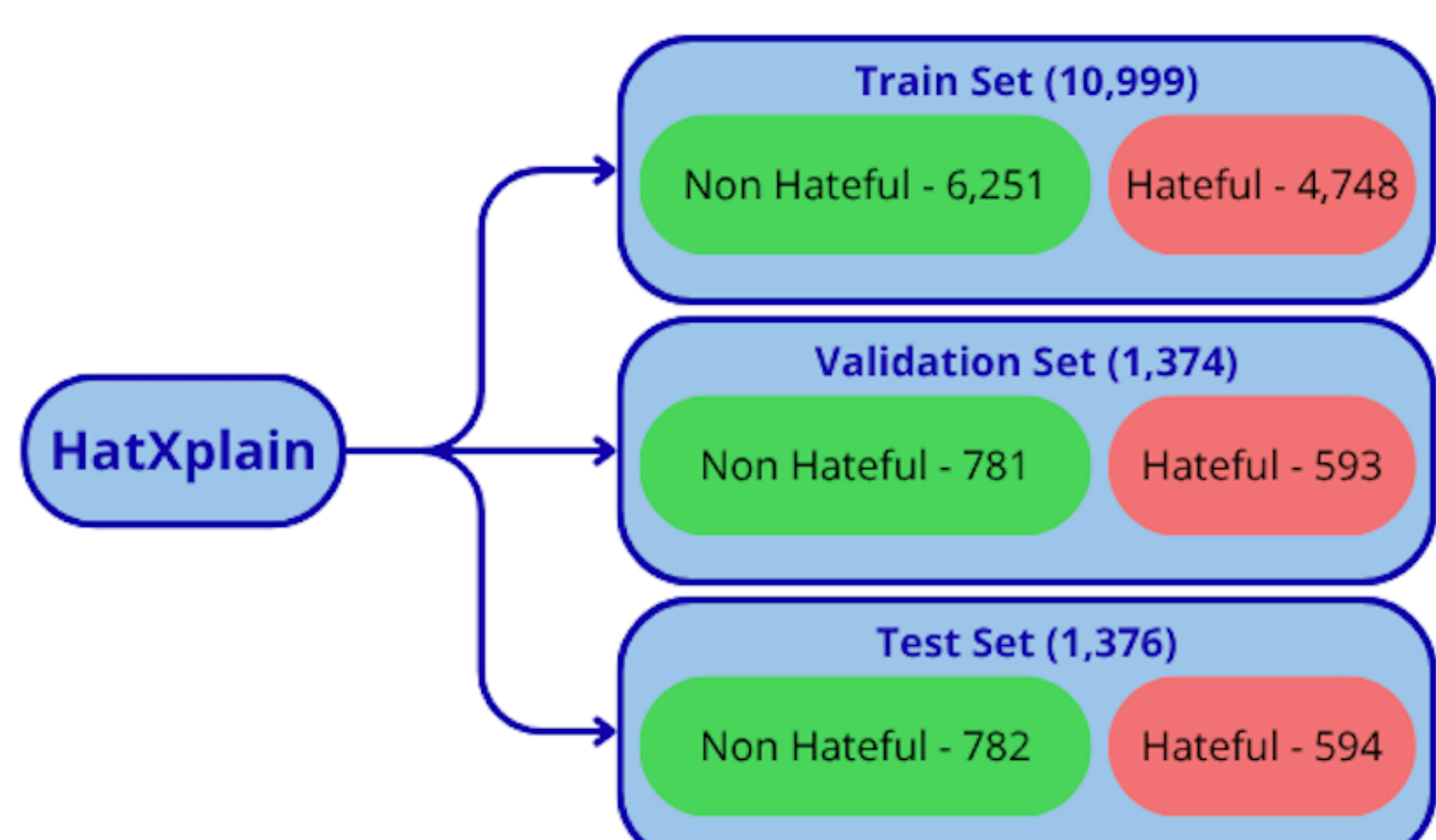
$$L = L_{CE}^{train} + \beta L_{CE}^{pairs} + \alpha L_{contrastive}$$

$$L_{contrastive} = \max(0, \text{margin} - \|z_{non-hateful} - z_{hateful}\|_2)^2$$

Dataset(s)

HateXplain (processed for binary classification) Original labels mapped to binary:

- ▶ hatespeech → hateful
- ▶ normal → non-hateful
- ▶ offensive → excluded



Validation

Contrastive Pair Creation

Baseline Validation False Positive

71

Kept as contrastive candidate

64

x2

Two rewrites per false positive add variation without changing the targeted identity cue.

Contrastive Pair Dataset

Non-hateful examples (from validation FPs)

128

Hateful counterparts (manually written)

128

Hyperparameter sweep - Contrastive grid search

Margin

{1,5,10,15,18,20,22}

Alpha

[0.05, 5.0]

Beta

[0.0, 1.0]

208

Unique Settings

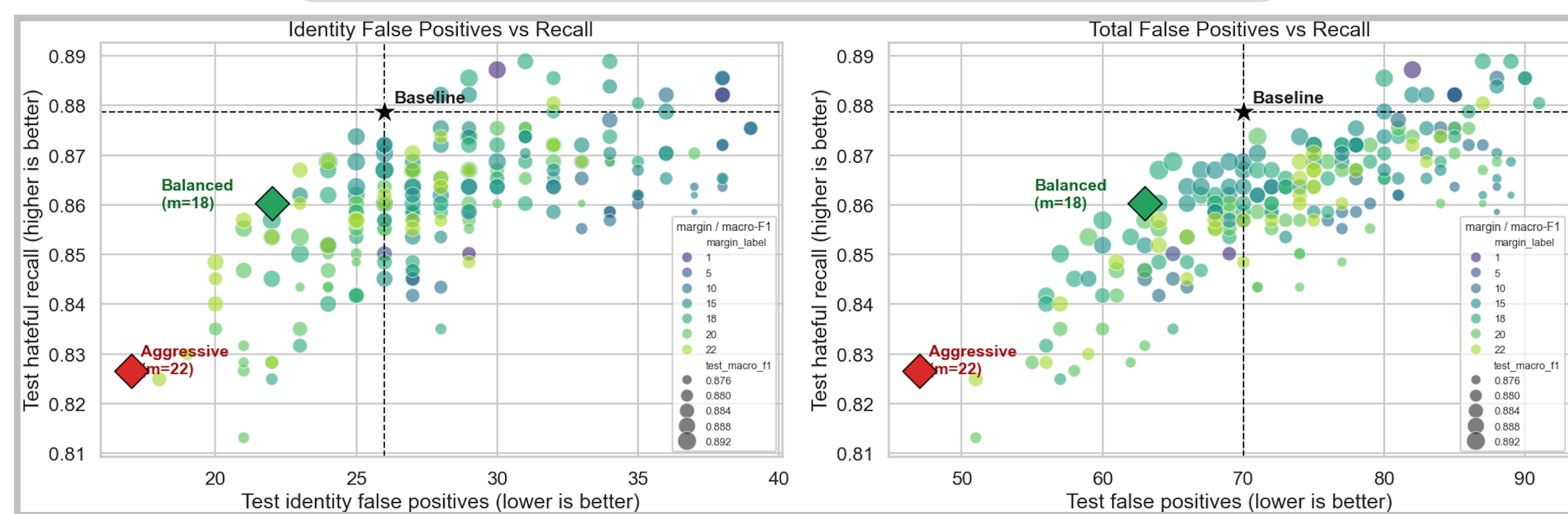
Test-Set Trade-Off: Fewer Identity False Positives vs Recall

Model	ID-FP	Recall	Macro-F1
Baseline	26	0.879	0.895
Balanced	22 (-4)	0.860 (-0.019)	0.891 (-0.003)
Aggressive	17 (-9)	0.827 (-0.052)	0.887 (-0.007)

Balanced → $\alpha = 0.2, \beta = 0.05, m = 18$

Aggressive → $\alpha = 2.0, \beta = 0.5, m = 22$

Trade-off Between Identity False Positives and Hateful Recall



Hyperparameter Effect

- ▶ **Margin** sets the minimum distance expected between paired hateful and non-hateful sentence embeddings.
- ▶ α controls how strongly the model is pushed to separate the paired embeddings.
- ▶ β controls how much the manual pairs contribute to the classification objective.

Stronger contrastive pressure makes the classifier more conservative: it predicts hate less often, which reduces identity false positives, but can also increase false negatives and lower hateful recall.

Margin ↑ / α ↑ / β ↑

Contrastive pressure ↑

Model separates paired hateful/non-hateful examples more

Identity false positives ↓

Hateful recall can also ↓

Limitations and Ethical Consideration

- ▶ The identity seed list is manually defined and incomplete.
- ▶ Integrated Gradients is diagnostic, not proof of causal model reasoning.
- ▶ Manually written contrastive examples may introduce style bias or unintended shortcuts.
- ▶ Binary classification simplifies the distinction between hate speech, offensive language, counterspeech, satire, and criticism.
- ▶ Labels come from few annotators per post, so ground truth may contain subjective or cultural bias.
- ▶ Hate speech data and generated counterparts require careful handling and review.
- ▶ The model lacks conversational, speaker, and platform context.

Conclusion

Model-error-driven contrastive fine-tuning reduces identity-related false positives, but creates a recall trade-off.

- ▶ The **balanced setting** reduced test identity false positives from 26 to 22, with only a small drop in recall and macro-F1.
- ▶ The **aggressive setting** reduced identity false positives further, from 26 to 17, but recall dropped more clearly. This shows that stronger contrastive pressure makes the classifier more conservative: it predicts hate less often, reducing false positives but missing more truly hateful examples.

Overall, the method acts as a control knob: stronger contrastive pressure makes the classifier more conservative, reducing over-flagging of identity-related non-hateful text while increasing the risk of missing truly hateful examples.

This shows that targeted contrastive pairs are useful for bias mitigation, but the final model must be selected according to the desired moderation trade-off.

References

- [1] S. Mathew, P. Saha, S. Yimam, C. Biemann, P. Goyal, and A. Mukherjee, "HateXplain: A Benchmark Dataset for Explainable Hate Speech Detection," in Proceedings of the AAAI Conference on Artificial Intelligence, vol. 35, no. 17, pp. 14867–14875, 2021.
- [2] I. Sen, M. Samory, C. Wagner, and S. Aref, "Counterfactually Augmented Data and Unintended Bias: The Case of Sexism and Hate Speech Detection," in Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), pp. 4716–4728, 2022.
- [3] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "RoBERTa: A Robustly Optimized BERT Pretraining Approach," arXiv preprint arXiv:1907.11692, 2019.